# LING324 Cognitive Linguistics

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Lecture 5: Mental Spaces

Lancaster University

# Overview

- Introduction
- Mental spaces
  - Modals
  - Semantic anomalies
  - Referential ambiguities
  - Change predicates
  - Tense
  - Conditionals
  - Epistemic distance
- Conceptual blending
  - Analogy
  - Counterfactual disanalogy
  - XYZ constructions
  - Non-linguistic examples

# Introduction

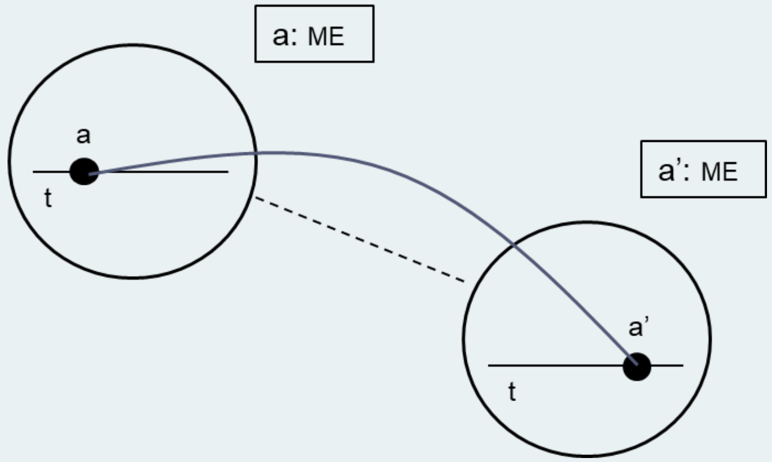
- Focus on processes of **meaning construction** in discourse
- Developed in opposition to referential or truth-conditional theories of semantics
- Resolves traditional semantic problems relating, e.g., to **reference, analogy, conditionality** and **counterfactuality**

a. I am getting ahead of myself

# Mental Spaces

- Words do not refer directly to real-world entities but to **elements** in a **mental space**

> Mental spaces are small conceptual packets constructed as we think and talk, for purposes of local understanding and action. They are interconnected, and can be modified as thought and discourse unfold (Fauconnier & Turner 1996: 113)



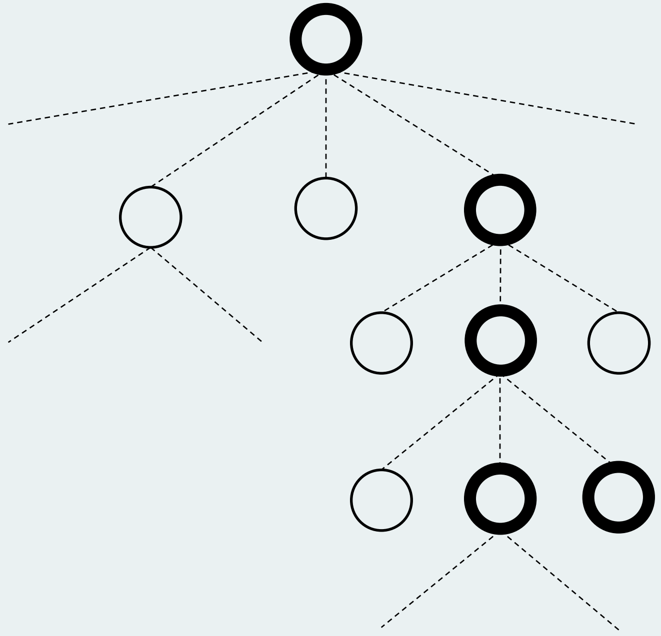

- Meaning construction involves:
  - The **building** of mental spaces
  - **Mappings** between elements in mental spaces

# Space Builders

- As discourse unfolds, new spaces established relative to **base space**
- Established by **space builders**:
  - Prepositional phrases *(during the last century)*
  - Adverbial phrases *(hypothetically)*
  - Modal verbs *(may, must)*
  - Conditional conjunctions *(if … then …)*
  - Subject-verb combination followed by embedded sentence *(Ivy thinks …)*
- Spaces populated by **elements** – supplied in discourse by NPs
- Structured internally by **frames** and other **ICMs**

# Counterparts and Connectors

- Elements assigned **properties** and enter into **relations** with other elements in same space
- Across spaces, elements linked to **counterpart elements** by **connectors**
- Counterparts established on basis of **pragmatic function: role, identity, representation**
- Produces complex **lattice** of spaces which discourse participants ‘move through’ with shifting **viewpoint** and **focus**



# Access Principle

- "An expression that names or describes an element in one mental space can be used to *access* a counterpart of that element in another space" (Fauconnier 1997: 41)

> **Access Principle**
>
> If two elements a and b are linked by a connector *F* (b = F(a)), then element b can be identified by naming, describing or pointing to its counterpart a

(Fauconnier 1997: 41)

- Element named or described = trigger; element accessed = target

# Context and Ambiguity

- Operations guided by context

    b. If I were your father, I would smack you

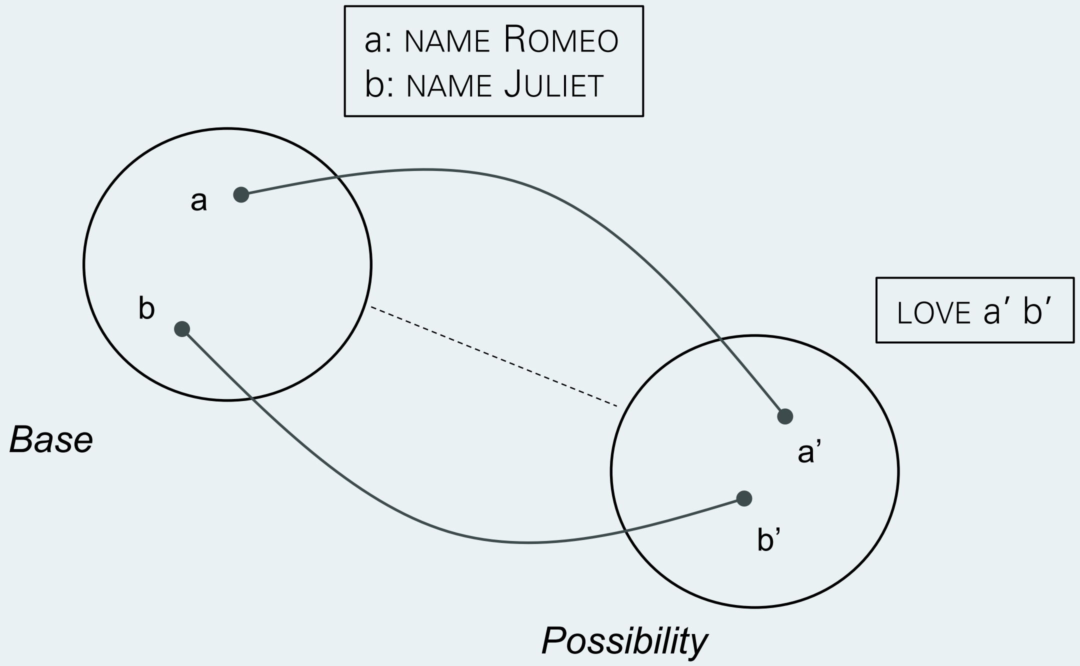
- Different interpretations based on different mappings and relations in counterfactual space
    - **Lenient father:** father inherits baby-sitter’s predisposition
    - **Stern father:** baby-sitter replaced by father
    - **Role:** baby-sitter is father



(Fauconnier 1997: 14-16)

# Modals

c. Maybe Romeo is in love with Juliet



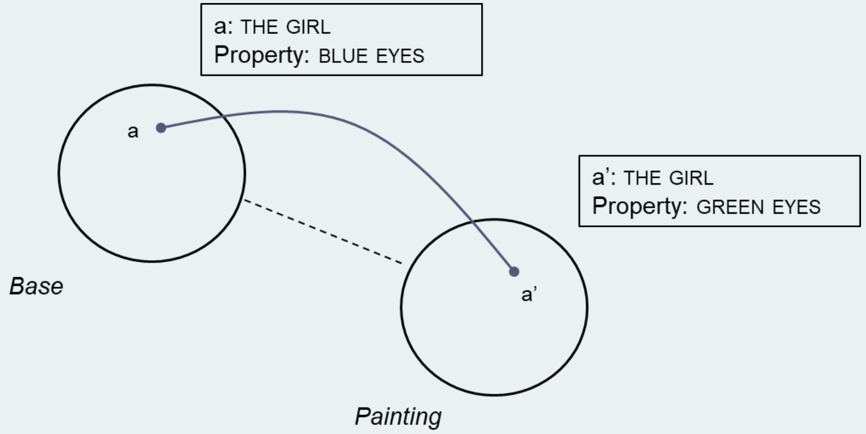



(Fauconnier 1997: 42-43)

# Semantic Anomalies

- Inherent contradiction resolved by mental spaces
    - Trigger = girl in real world
    - Target = girl in painting
    - Pragmatic function = representation

d. In the painting, the girl with blue eye has green eyes



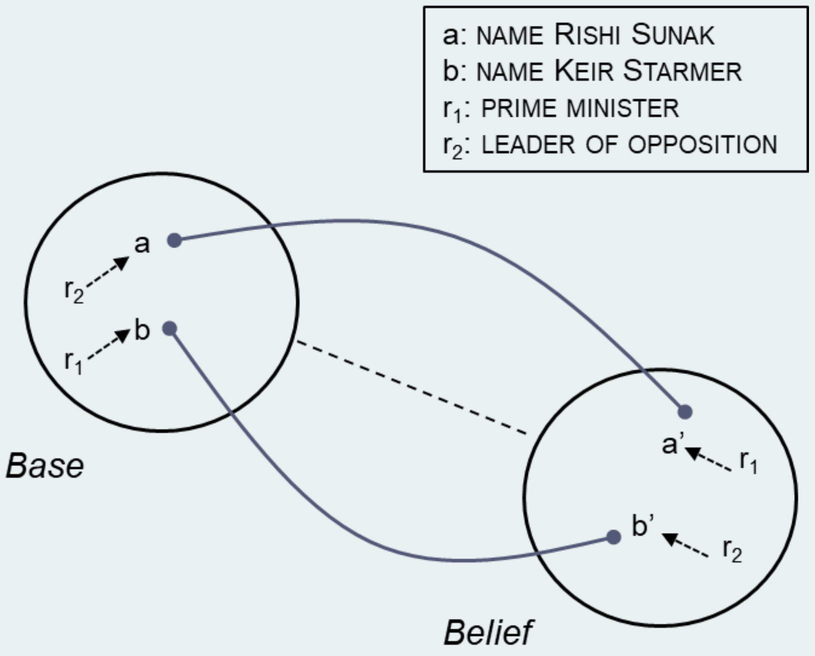

(Fauconnier 1994: 10ff)

# Referential Ambiguities

e. The Prime Minister changes every five years

- Ambiguous between **role** and **value** reading (Fauconnier 1994: 39ff)
- Pragmatic function **role** vs. **identity**

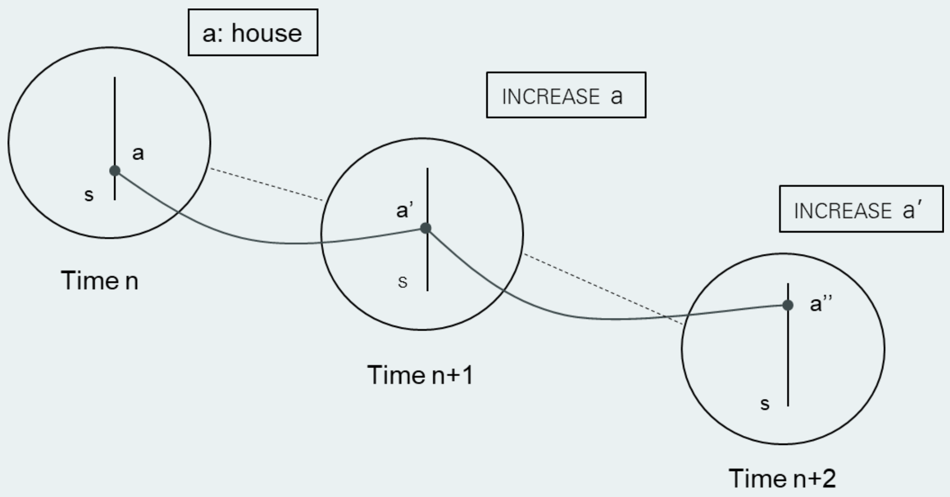
f. Rishi Sunak thinks he's still the Prime Minister and Kier Starmer is the leader of the opposition



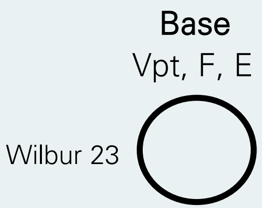
# Change Predicates

g. Ivy's house keeps getting bigger

- *Keep* is space builder: establishes series of chronologically ordered spaces
- Normal interpretation: NP = same entity
- Variable interpretation: NP = range of different entities



# Tense




- Function of tense is **discourse management**: enables hearer to track events in time
- Guides hearer through network of spaces shifting **viewpoint** and **focus**
  - Viewpoint = space from which other spaces currently built or accessed
  - Focus = space where content currently being added
  - Event = time associated with event described (often same as focus)

h. Wilbur is twenty-three. He has lived abroad. In 1990, he lived in Rome. In 1991 he would move to Venice. He would then have lived a year in Rome.

(Fauconnier 1997:73-75)

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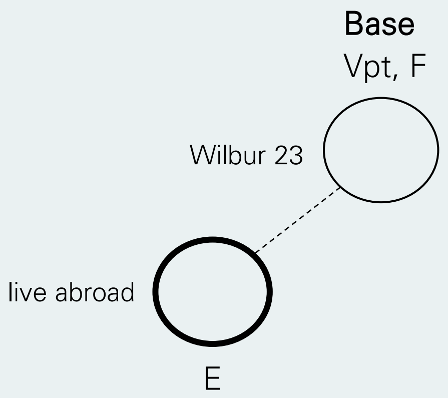


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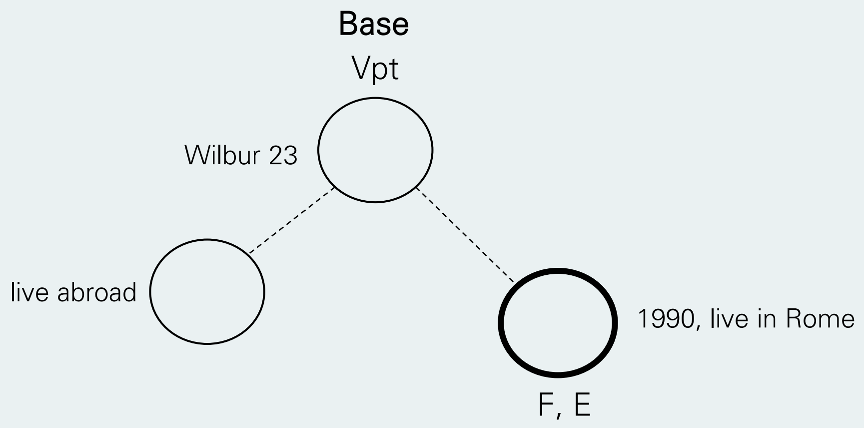


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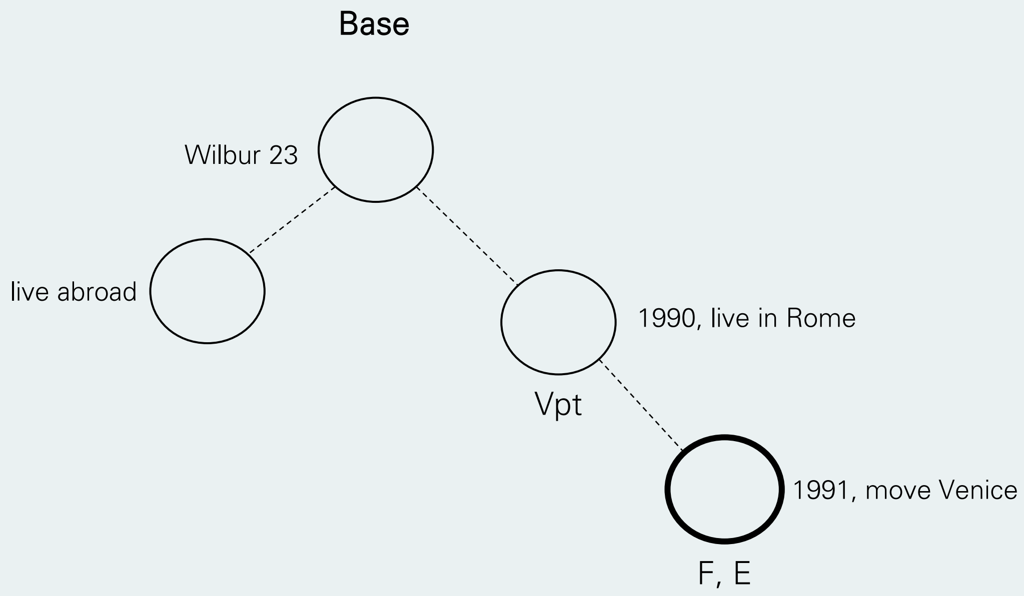


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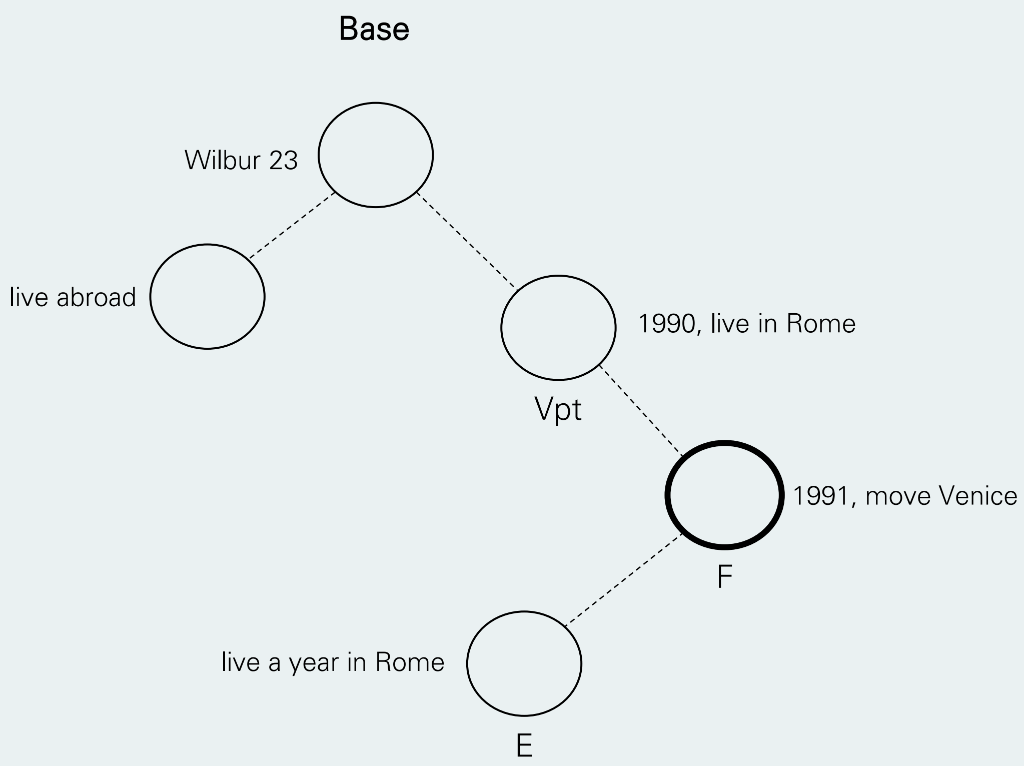


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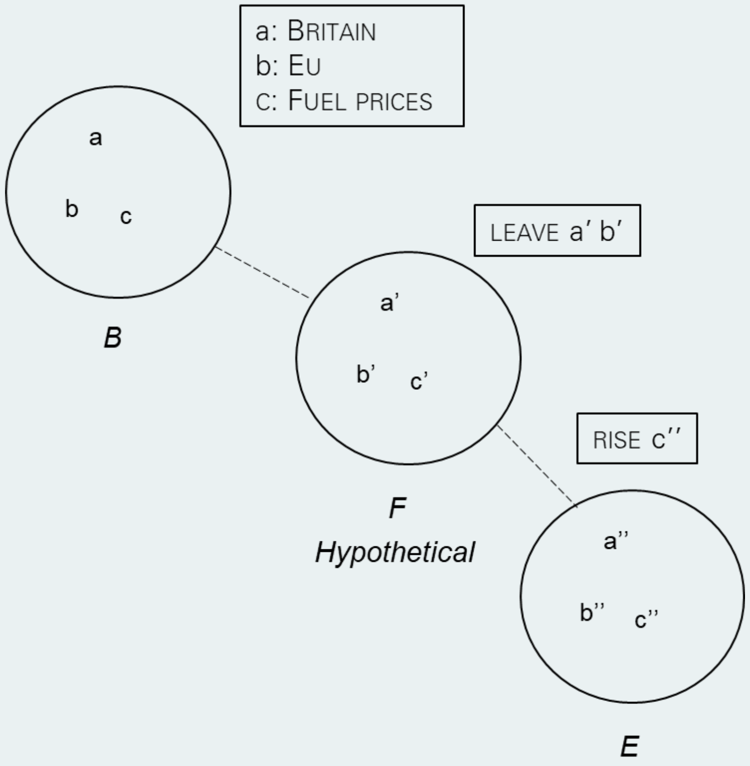
# Conditionals

- Form of 'If A then B'
- A = antecedent; B = consequent
- **Foundation** (F) and **Expansion** (E) spaces (Fauconnier 1997: 131-138)
- Foundation **hypothetical** or **counterfactual** relative to base

   i. If Britain leaves the EU, then fuel prices will rise

   j. If Britain had remained in the EU, then fuel prices would not have risen



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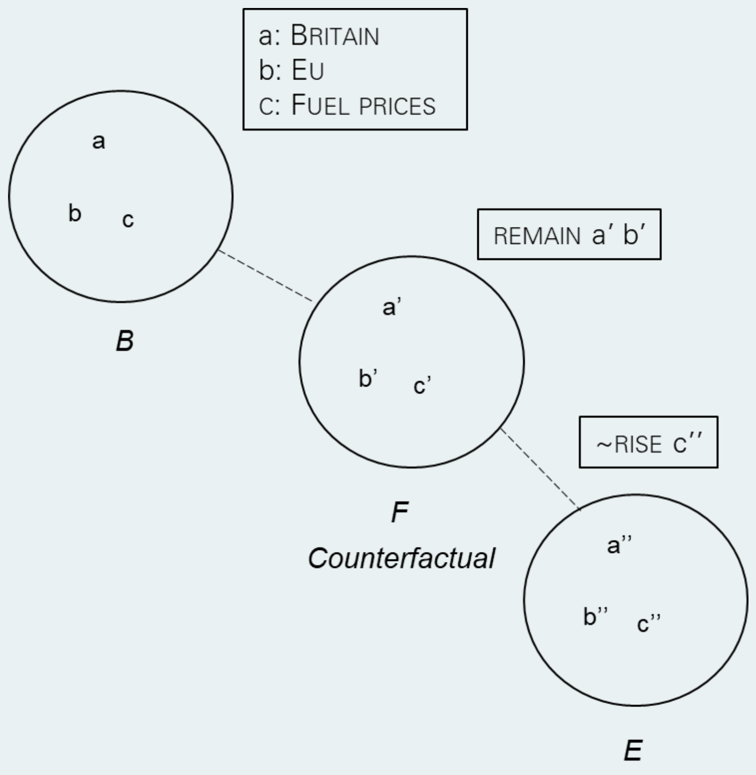


# Epistemic Distance

- Tense used to mark **epistemic distance** in conditional constructions
  - degree to which speaker assesses 'hypothesis' in *F* as likely or unlikely to hold in reality space (Fauconnier 1997: 93)
- Past tense marks greater epistemic distance
- Spreads to all spaces subordinate to hypothetical space (Sweetser 1996)

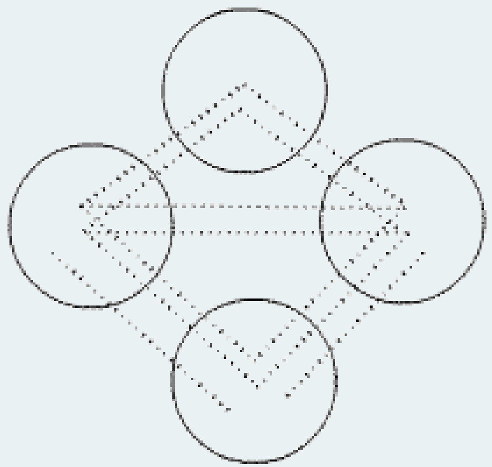
k. If Britain leaves the EU, then fuel prices will rise

l. If Britain left the EU, then fuel prices would rise

# Conceptual Blending

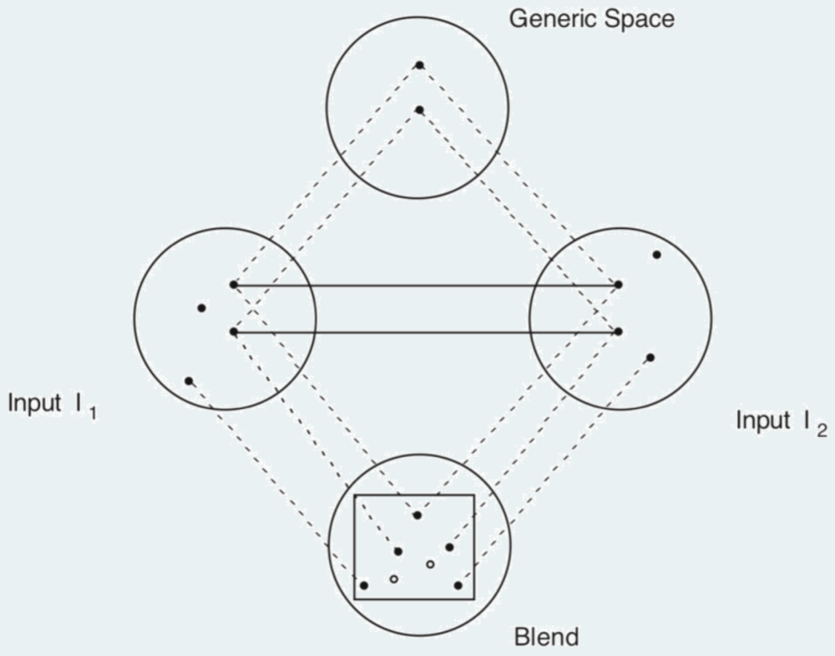
- Mental spaces enter into **conceptual integration networks** to create **blends** (Fauconnier & Turner 2002)
- Implicated in creative aspects of meaning construction like **novel metaphor**
- Pervasive in human imagination, reason and language

> Blends abound in all kinds of cases that go largely unnoticed. Some are created as we talk, others are conventional, and others are even more firmly entrenched in the grammatical structure. (Fauconnier 1997: 155)



# Conceptual Blending

- Four-space model of online meaning construction
- **Mappings** and **projections**
- Input spaces structured by **frames**
- **Vital relations** held between **counterpart elements**
    - change; identity; time; space; cause-effect; part-whole; representation; role; analogy; disanalogy; property; similarity; category; intentionality; uniqueness
- **Emergent structure** through **composition, completion** and **elaboration**



# Analogy

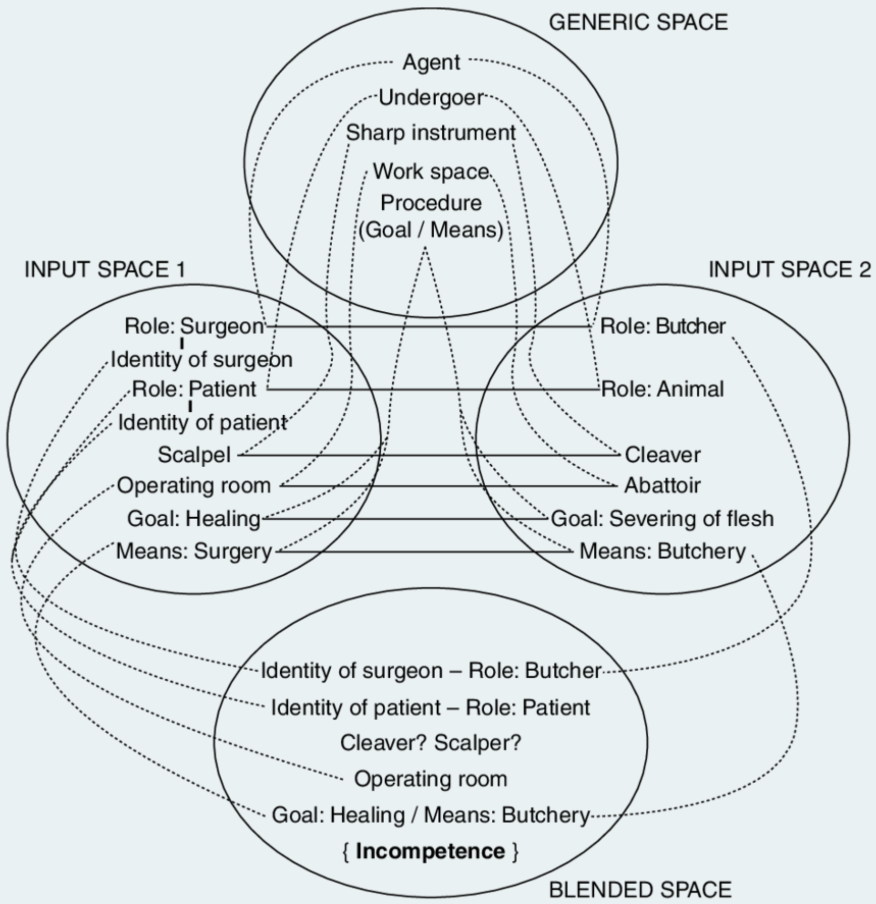
- Metaphors not accounted for by unidirectional mapping (CMT)

  m. This surgeon is a butcher

  Inference: The surgeon is incompetent

- **Double-scope network:** frame-level structure projected from both input spaces

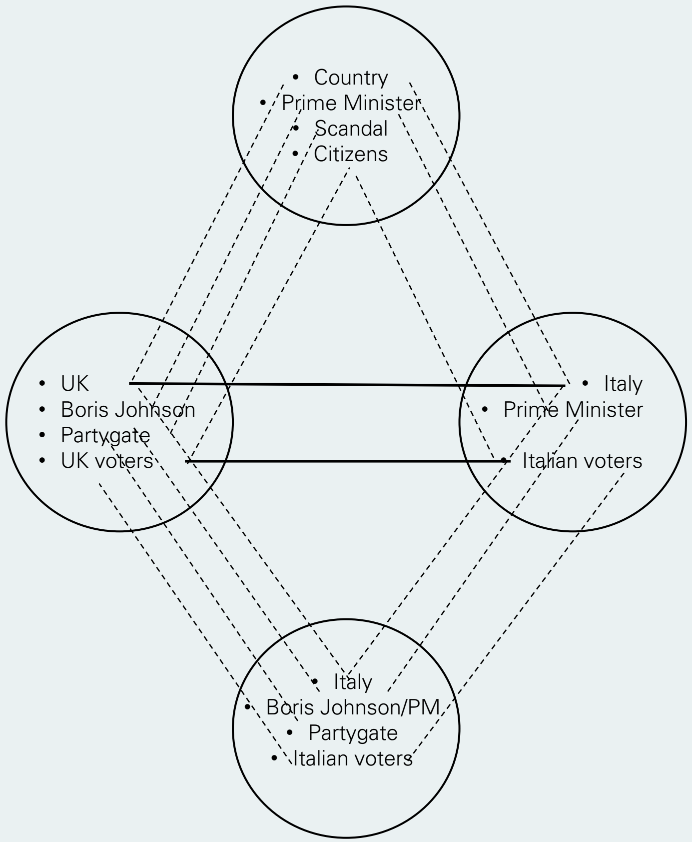
In the blended space, the means of butchery have been combined with the ends, the individuals and the surgical context of the surgery space. The incongruity of the butcher's means with the surgeon's ends leads to the central inference that the butcher is incompetent. (Grady, Oakley & Coulson 1999: 105)



# Counterfactual Disanalogy

n. In Italy, Partygate wouldn't have done Boris Johnson any harm

- Highlights dissimilarity between two contexts: ITALIAN POLITICS and UK POLITICS
- Disanalogy not explained by standard MST
- Obtained based on integration of two spaces:
    - Partial projection resulting in blended structure
    - Imagined scenario in which Johnson is Prime Minister of Italy and Partygate took place in Italian context
    - Back projection highlighting contrast between Inputs

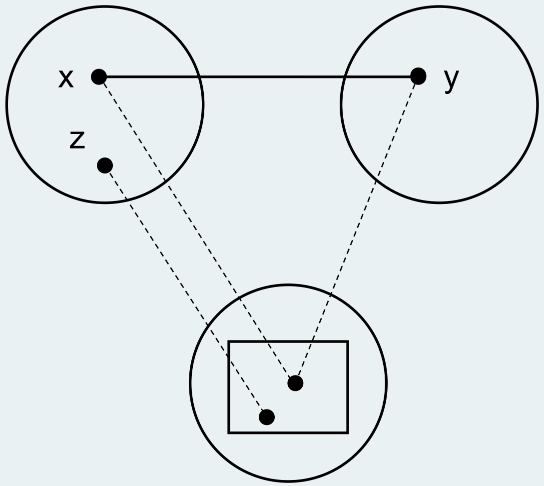



Emergent structure: Partygate doesn't harm Johnson as PM of Italy
Inference: disanalogy between UK and Italy, UK voters and French voters, attitudes to politicians and scandal, etc.

# XYZ Constructions

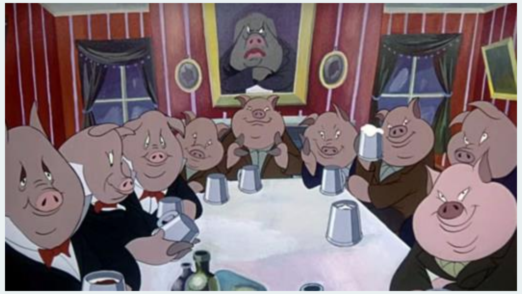
- X be Y of Z
- A grammatical **construction** “whose purpose is to systematically prompt for blends” (Fauconnier & Tuner 2002: 142)

  - Money is the root of all evil
  - Religion is the opiate of the masses
  - Death is the mother of beauty
  - The Pope is the father of the Catholic Church



# Non-linguistic Examples

- Blending independent of language
- Operation fundamental to human thinking
- Manifest in non-linguistic forms of imagination/creativity
- E.g. **anthropomorphism** (Turner 1996)



# Thank you

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